

Jungwoo Lee

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Master's student in Computer Science with 12+ years of hands-on experience spanning Software Engineering, Software Development, Database Administration, and Middleware Administration. Specialized in building scalable and high-efficiency solutions using REST APIs, Java, Python, and C programming. Demonstrated expertise in managing large-scale transactional databases, maintaining mission-critical systems, and optimizing middleware performance. Passionate about applying a unique blend of academic knowledge and professional experience to deliver transformative solutions in mission-critical systems.

EDUCATION

M.S. Computer Science	May 2025 (Expected)
Arizona State University, Tempe, AZ	GPA: 3.78/4.0
Relevant Coursework: Statistical Machine Learning, Real-Time Embedded Systems, Software Security	
B.S. Computer Engineering	Feb 2011
Kyungpook National University, Daegu, South Korea	GPA: 3.89/4.3 (Rank: 1/16)
University of Oklahoma, Norman, OK (Exchange Student, Aug 2009-May 2010)	GPA: 4.0/4.0

PROFESSIONAL EXPERIENCE

Korea Financial Telecommunications & Clearings Institute (KFTC)	Seoul, South Korea
Principal Software Engineer	Feb 2017–Present
● Real-time loan transfer system • Project lead and developer of a high-performance gateway system using Java, Apache Camel, and REST APIs, facilitating seamless data exchange between 50+ financial institutions. • Ensured stability and accuracy under high-load conditions, processing 230,000+ loans worth \$10.6B., and reducing interest rates by 1.52 percentage points. • Demonstrated exceptional communication skills by collaborating with business stakeholders. • Trained 3+ new developers on system features. • Recognized with a citation by the Financial Services Commission. ● Financial account management platform • Engineered and developed a platform leveraging Java, C, Apache Camel, REST APIs to provide real-time access to financial assets across 60+ institutions, recovering \$2.9B in forgotten assets. • Developed and maintained a web application using Spring Framework for internal users, utilizing the MVC architecture. • Built and maintained a C-based batch program to reconcile daily transactions. ● One-Stop account freeze system • Contributed as a team member to the development of a fraud prevention backend application utilizing Java and Apache Camel, enabling consumers to freeze accounts across institutions. • Collaborated with financial institutions and regulatory bodies to process 490,000+ transactions per year.	
Senior Database and Middleware Administrator	Aug 2012–Jan 2017
• Key contributor in managing large-scale transaction processing for critical financial systems, ensuring high availability, data integrity, and zero downtime. • Led database optimization efforts through SQL tuning, indexing, and partitioning, improving system performance and reliability.	
Process Engineer	Jan 2011–Jul 2012
• Played a key role in IT Service Management (ITSM) initiatives, managing outsourcing teams to enhance service quality and securing ISO 20000 certification, which led to a 20% reduction in annual operating costs.	

Republic of Korea Army

Sergeant

- Deployed to Iraq, contributing to peacekeeping, reconstruction efforts and multinational military cooperation.

South Korea / Iraq

Sep 2004–Sep 2006

PROJECTS

- Optimizing Recommendation Systems with Large Language Models, *Class Team Project*

Spring 2024

- Developed a hybrid recommendation system integrating collaborative filtering with GPT and LLaMA models, enhancing recommendation accuracy by 20%.
- Real-Time Embedded Systems for Autonomous Robotics, *Class Team Project*

Fall 2023

- Developed embedded systems in C and Lingua Franca for real-time sensor integration and motor control.
- AI-Generated Scientific Paper Detection, *Class Team Project*

Fall 2023

- Designed a machine learning model to classify AI-generated and human-written scientific papers to achieve high F1 scores with optimization of TF-IDF and parameter-tuning

CERTIFICATION

- ITIL V3 Foundation, EXIN

Nov 2011
- Engineer Information Processing, Human Resources Development Service of Korea

Aug 2010

TECHNICAL SKILLS

Programming Languages: Java, SQL, C, C++, Python

Frameworks and Libraries: Spring, Apache Camel, JPA(Hibernate), Scikit-learn, PyTorch, PySpark, Hugging Face

Tools and Technologies: REST API, Shell scripting, JEUS, Apache Tomcat, Git, Bitbucket, Confluence

Databases: Oracle, PostgreSQL, SQLite, Tiberio

Operating Systems: UNIX(Solaris, HP-UX), Linux(Red Hat, Ubuntu), Windows Server